

Paper Practice 1.1 • WORKED KEY**Numerical Expressions**

Show all of your work in the space provided. No calculator. For translation problems, write the expression FIRST, then evaluate it. Circle your final answers.

- 1.** Simplify each expression. (Warm-Up Review)

a. $-3 + (-14)$

Answer: -17

b. $15 - 22 + (-6)$

Answer: $15 - 22 = -7$; $-7 + (-6) = -13$

c. $-9 - (-5) - (-3)$

Answer: $-9 + 5 + 3 = -1$

- 2.** Write a numerical expression for each phrase.
Do NOT evaluate. (Translate)

a. four plus twelve divided by six

Answer: $4 + 12 \div 6$

b. six times eleven minus three times seven

Answer: $6 \cdot 11 - 3 \cdot 7$

c. fifteen less than three times eight

Answer: $3 \cdot 8 - 15$ ("less than" REVERSES the order – not $15 - 3 \cdot 8$)

- 3.** Write a numerical expression for each phrase, using grouping symbols where they are needed. Then evaluate. (Translate)

a. the sum of seven and eleven, divided by three

Answer: $(7 + 11) \div 3 = 18 \div 3 = 6$

b. five times the difference of twelve and four

Answer: $5(12 - 4) = 5(8) = 40$

c. twice the sum of nine and six

Answer: $2(9 + 6) = 2(15) = 30$

- 4.** Write and evaluate a numerical expression for EACH phrase below, then explain in one sentence why the two answers are different. (Compare)

a. the sum of eight and four, divided by two

Answer: $(8 + 4) \div 2 = 12 \div 2 = 6$

b. eight plus four divided by two

Answer: $8 + 4 \div 2 = 8 + 2 = 10$. In (a) the words "the sum of" group $8 + 4$ so it is done first; in (b) there is no grouping, so division happens before addition.

5. A bowling handicap is found by subtracting the player's average from 220, multiplying that result by 3, and then dividing by 4. Write a numerical expression for the handicap of a player whose average is 180, then evaluate it. (Write & Evaluate)

Answer: $3(220 - 180) \div 4 = 3(40) \div 4 = 120 \div 4 = 30$

6. Answer each part. (Powers)

- a. Write $5 \cdot 5 \cdot 5 \cdot 5$ using an exponent, then name the base and the exponent.

Answer: 5^4 – base 5, exponent 4

- b. Evaluate 3^4

Answer: $3 \cdot 3 \cdot 3 \cdot 3 = 81$

- c. Evaluate 2^6

Answer: $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 64$

10. Evaluate: $(3^3 - 7) \div (12 + 8)$ (GEMS)

Answer: the fraction bar groups the whole top and the whole bottom: $(27 - 7) \div (12 + 8) = 20 \div 20 = 1$

11. Error Analysis – Gus evaluated the expression below and wrote 16. Identify his mistake, then evaluate correctly. (Error Analysis)

Gus's work: $4^2 - 6 \div 2 + 3 = 16 - 3 + 3 = 16$

Answer: He ignored the fraction bar, which is a GROUPING symbol – it groups the entire numerator and the entire denominator. Correct: $(4^2 - 6) \div (2 + 3) = (16 - 6) \div 5 = 10 \div 5 = 2$

7. Evaluate: $30 - 4 + 5^2 - 6 \cdot 9$ (GEMS)

Answer: exponent first: $30 - 4 + 25 - 54$; then multiply: $6 \cdot 9 = 54$; left to right: $26 + 25 = 51$, $51 - 54 = -3$

8. Evaluate: $(8 + 4)^2 \div 4(9 - 3)$ (GEMS)

Answer: grouping first: $(12)^2 = 144$ on top; $4(6) = 24$ on bottom; $144 \div 24 = 6$

9. Evaluate: $20 - [8 + (5 - 3)^2] + 7$ (GEMS)

Answer: innermost first: $(5 - 3)^2 = 2^2 = 4$; $[8 + 4] = 12$; $20 - 12 + 7 = 15$

12. Mr. Nappier's class is planning a field trip for 24 students. The costs are shown in the table. Write a numerical expression for the total cost of the bus, admission, and lunch for every student, then evaluate it. If the class budget is \$650, can they afford it? Explain. (Write & Evaluate)

Item	Cost
Bus (flat fee)	\$180
Admission	\$12 per student
Lunch	\$8 per student
T-shirt	\$6 per student

Answer: $180 + 12(24) + 8(24) = 180 + 288 + 192 = \660 . NO – the trip costs \$660, which is \$10 OVER the \$650 budget. (T-shirts are not part of this expression.)